

nRF54L15

Revision 2

Errata

v1.1

Contents

1	nRF54L15 Revision 2 Errata	3
2	Revision history	4
3	New and inherited anomalies	6
3.1	[1] CCM: MAC error is returned if MDATA size is slightly larger than MLEN.	7
3.2	[6] RADIO: Spectrum of transmitted signal is not optimal for 802.15.4.	7
3.3	[7] UARTE: RXD.AMOUNT is not updated after RXFLUSH if FIFO is empty.	8
3.4	[8] SPIM: Wrong data is transmitted on MOSI.	9
3.5	[10] SPU: PERM and FEATURE registers cannot be locked.	10
3.6	[13] KMU: Provision task is not ignored when KMU is busy.	10
3.7	[16] VPR: Cleared EVENTS are triggered again on the next CNT0 event.	11
3.8	[17] VPR: Bus faults during hibernate save sequence are ignored.	11
3.9	[18] VPR: Subsequent VPR hibernate sleeps might fail to save ra.	12
3.10	[20] CLOCK: RADIO payload is not transmitted.	12
3.11	[21] SPIM: MOSI line toggles after transmission ends.	13
3.12	[22] TIMER: EVENTS_COMPARE[n] event is not generated.	13
3.13	[23] RRAMC: READY and READYNEXT events are generated when CPU wakes up.	14
3.14	[24] TIMER: EVENT_COMPARE is not generated when using PPI.	14
3.15	[25] TIMER: EVENT_COMPARE[n] is not generated.	15
3.16	[26] DPPI: PPIB between domains uses wrong security attribution.	15
3.17	[30] CLOCK, GRTC: GRTC operates incorrectly at low temperature.	16
3.18	[37] POWER: Current consumption might increase after pin reset or power cycle.	16
3.19	[39] CLOCK: Device can behave erratically after XOSTART.	17
3.20	[41] RADIO: Sensitivity is degraded on n*32 MHz channels.	17
3.21	[44] EASYDMA: DMA.RX.AMOUNT is not updated with bytes count after match event.	18
3.22	[47] POWER: Device resets while debugging in System OFF mode.	18
3.23	[48] RESET: RESETREAS register might show an additional wakeup source after reset.	18
3.24	[49] RADIO: First bits of on-air packet are not correct.	19
3.25	[50] RRAMC: SPU00 incorrectly reports that RRAMC has DMA.	20
3.26	[54] SPIS: SDO is configured as an input when SPIS is idle.	20
3.27	[58] COMP: Static 300 µA current after COMP has been enabled.	21
3.28	[59] EGU: EGU uses incorrect security attribution.	21
3.29	[63] RESET: Soft reset might not fully reset the device.	22
3.30	[100] GPIO: Pins with analog functionality might be shorted together during power-on reset.	22
3.31	[102] CCM: Operation fails when CCM MAC length is set to 0.	23
3.32	[103] GPIO: State is inconsistent during power-on reset.	23
3.33	[105] TWIM: Peripheral becomes unresponsive when disabled during clock stretching.	24
3.34	[111] SAADC: Output data is incorrect when noise shaping is used.	24
3.35	[114] GPIO: Current consumption increases when Wake on pin is used.	25
4	Fixed anomalies	26
5	New and inherited configurations	27
5.1	[56] RADIO: Regulatory limits might be exceeded when using external FEM or PA.	27

1 nRF54L15 Revision 2 Errata

This Errata document contains anomalies and configurations for the nRF54L15 SoC, Revision 2 (QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00).

The document indicates which anomalies are fixed, inherited, or new compared to [Revision 1](#).

2 Revision history

See the following list for an overview of changes from previous versions of this document.

Version	Date	Change
nRF54L15 Revision 2 v1.1	12.05.2026	• Added: No. 10. "PERM and FEATURE registers cannot be locked" • Added: No. 13. "Provision task is not ignored when KMU is busy" • Added: No. 50. "SPU00 incorrectly reports that RRAMC has DMA" • Added: No. 54. "SDO is configured as an input when SPIS is idle " • Added: No. 56. "Regulatory limits might be exceeded when using external FEM or PA" • Added: No. 58. "Static 300 μA current after COMP has been enabled" • Added: No. 59. "EGU uses incorrect security attribution" • Added: No. 63. "Soft reset might not fully reset the device" • Added: No. 100. "Pins with analog functionality might be shorted together during power-on reset" • Added: No. 102. "Operation fails when CCM MAC length is set to 0" • Added: No. 103. "State is inconsistent during power-on reset" • Added: No. 105. "Peripheral becomes unresponsive when disabled during clock stretching" • Added: No. 111. "Output data is incorrect when noise shaping is used" • Added: No. 114. "Current consumption increases when Wake on pin is used"

Version	Date	Change
nRF54L15 Revision 2 v1.0	18.06.2025	- Added: No. 1. "MAC error is returned if MDATA size is slightly larger than MLEN" - Added: No. 6. "Spectrum of transmitted signal is not optimal for 802.15.4" - Added: No. 7. "RXD.AMOUNT is not updated after RXFLUSH if FIFO is empty" - Added: No. 8. "Wrong data is transmitted on MOSI" - Added: No. 16. "Cleared EVENTS are triggered again on the next CNT0 event" - Added: No. 17. "Bus faults during hibernate save sequence are ignored" - Added: No. 18. "Subsequent VPR hibernate sleeps might fail to save ra" - Added: No. 20. "RADIO payload is not transmitted" - Added: No. 21. "MOSI line toggles after transmission ends" - Added: No. 22. "EVENTS_COMPARE[n] event is not generated" - Added: No. 23. "READY and READYNEXT events are generated when CPU wakes up" - Added: No. 24. "EVENT_COMPARE is not generated when using PPI" - Added: No. 25. "EVENT_COMPARE[n] is not generated" - Added: No. 26. "PPIB between domains uses wrong security attribution" - Added: No. 30. "GRTC operates incorrectly at low temperature" - Added: No. 37. "Current consumption might increase after pin reset or power cycle" - Added: No. 39. "Device can behave erratically after XOSTART" - Added: No. 41. "Sensitivity is degraded on n*32 MHz channels" - Added: No. 44. "DMA.RX.AMOUNT is not updated with bytes count after match event" - Added: No. 47. "Device resets while debugging in System OFF mode" - Added: No. 48. "RESETREAS register might show an additional wakeup source after reset" - Added: No. 49. "First bits of on-air packet are not correct"

3 New and inherited anomalies

The following anomalies are present in Revision 2 of the nRF54L15 SoC.

ID	Module	Description	Inherited from Revision 1
1	CCM	MAC error is returned if MDATA size is slightly larger than MLEN	X
6	RADIO	Spectrum of transmitted signal is not optimal for 802.15.4	X
7	UARTE	RXD.AMOUNT is not updated after RXFLUSH if FIFO is empty	X
8	SPIM	Wrong data is transmitted on MOSI	X
10	SPU	PERM and FEATURE registers cannot be locked	X
13	KMU	Provision task is not ignored when KMU is busy	X
16	VPR	Cleared EVENTS are triggered again on the next CNT0 event	X
17	VPR	Bus faults during hibernate save sequence are ignored	X
18	VPR	Subsequent VPR hibernate sleeps might fail to save ra	X
20	CLOCK	RADIO payload is not transmitted	X
21	SPIM	MOSI line toggles after transmission ends	X
22	TIMER	EVENTS_COMPARE[n] event is not generated	X
23	RRAMC	READY and READYNEXT events are generated when CPU wakes up	X
24	TIMER	EVENT_COMPARE is not generated when using PPI	X
25	TIMER	EVENT_COMPARE[n] is not generated	X
26	DPPI	PPIB between domains uses wrong security attribution	X
30	CLOCK, GRTC	GRTC operates incorrectly at low temperature	X
37	POWER	Current consumption might increase after pin reset or power cycle	X
39	CLOCK	Device can behave erratically after XOSTART	X
41	RADIO	Sensitivity is degraded on n*32 MHz channels	X
44	EASYDMA	DMA.RX.AMOUNT is not updated with bytes count after match event	X
47	POWER	Device resets while debugging in System OFF mode	X
48	RESET	RESETREAS register might show an additional wakeup source after reset	X
49	RADIO	First bits of on-air packet are not correct	X
50	RRAMC	SPU00 incorrectly reports that RRAMC has DMA	X
54	SPIS	SDO is configured as an input when SPIS is idle	X
58	COMP	Static 300 µA current after COMP has been enabled	X

ID	Module	Description	Inherited from Revision 1
59	EGU	EGU uses incorrect security attribution	X
63	RESET	Soft reset might not fully reset the device	X
100	GPIO	Pins with analog functionality might be shorted together during power-on reset	X
102	CCM	Operation fails when CCM MAC length is set to 0	X
103	GPIO	State is inconsistent during power-on reset	X
105	TWIM	Peripheral becomes unresponsive when disabled during clock stretching	X
111	SAADC	Output data is incorrect when noise shaping is used	X
114	GPIO	Current consumption increases when Wake on pin is used	X

Table 1: New and inherited anomalies

3.1 [1] CCM: MAC error is returned if MDATA size is slightly larger than MLEN

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

A MAC error is generated.

Conditions

The MDATA size is larger than MLEN by 1 or a few bytes.

Consequences

Payload is lost due to MAC error.

Workaround

Use MLEN or a significantly bigger value than MLEN for the size of the MDATA job. For example, use mdata.size=1000 when a payload length in the range from 1 to 251 bytes is expected, but the exact payload length is not known beforehand.

3.2 [6] RADIO: Spectrum of transmitted signal is not optimal for 802.15.4

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

The FCC restricted band edge requirement might not be met for 2405 MHz and 2480 MHz.

Conditions

The radio is operating in the IEEE 802.15.4 mode at TX output power level above 0 dBm.

Consequences

The limit for emissions outside of the band might be violated for 2405 MHz and 2480 MHz.

Workaround

Perform the following register write immediately after setting RADIO.MODE to ieee802154_250Kbit:

```
*(volatile uint32_t *) 0x5008A810 = 2;
```

The workaround can be applied to all channels and output powers.

3.3 [7] UARTE: RXD.AMOUNT is not updated after RXFLUSH if FIFO is empty

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

Wrong number of transferred bytes is reported.

Conditions

During the handling of the RXTO event, FLUSHRX is performed and the number of transferred bytes does not exceed MAXCNT.

Consequences

UARTE operations are unreliable.

Workaround

Apply the following workaround:

```
/* Setup buffer to which data can be flushed. */
UARTE->DMA.RX.PTR = buf;
UARTE->DMA.RX.MAXCNT = 5;
/* Clear READY. Needed for the workaround. */
UARTE->EVENTS_DMA.RX.READY = 0;
UARTE->EVENTS_DMA.RX.END = 0;
/* Trigger flushing. */
UARTE->TASKS_FLUSHRX = 1;
/* Wait until flushing is done. */
while (UARTE->EVENTS_DMA.RX.END == 0);
/* If READY event is set then use AMOUNT value else FIFO was empty.
If not for the PAN, AMOUNT could be read without checking READY event. */
fifo_cnt = (UARTE->EVENTS_DMA.RX.READY == 1) ? UARTE->DMA.RX.AMOUNT : 0;
```

This workaround clears the `EVENTS_DMA.RX.READY` before starting the flushing, and reads `EVENTS_DMA.RX.READY` after the flushing to determine if FIFO had data and `DMA.RX.AMOUNT` has valid data.

3.4 [8] SPIM: Wrong data is transmitted on MOSI

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

Wrong data is sampled.

Conditions

SPIM is configured with `CPHA` to 0, `PRESCALER` is larger than 2, and the first transmitted bit is 1.

Consequences

Wrong data is transmitted.

Workaround

Use the following workaround if `PRESCALER` is larger than 2.

Before triggering `SPIM.TASKS_START`, complete the following steps:

1. Set `SPIM.IFTIMING.CSNDUR` = `SPIM.PRESCALER / 2 + 1` or any larger value.
2. Write value 0x82 to the corresponding SPIM instance address offset 0xc84.

When `SPI.EVENTS_STARTED` has been generated, write value 0x00 to the corresponding SPIM instance address offset 0xc84.

3.5 [10] SPU: PERM and FEATURE registers cannot be locked

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

Peripheral security settings cannot be locked for the peripherals with split security features. User-selectable Arm® TrustZone® map cannot be locked for peripherals with split security features. When lock bit is set, secure code can change the peripheral and split security settings.

Conditions

PERM lock bit is set for configurable security peripherals. FEATURE lock bits are set for split security peripherals.

Consequences

The security settings cannot be locked.

Workaround

None. Ensure software running in a secure environment does not change the PERM or FEATURE registers.

3.6 [13] KMU: Provision task is not ignored when KMU is busy

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

An incorrect ERROR event is generated when the TASKS_PROVISION task is triggered while TASKS_REVOKE is in progress.

Conditions

TASKS_PROVISION is triggered with SRC register not holding a RAM address while TASKS_REVOKE is in progress.

Consequences

The ERROR event might be misattributed to the revoke operation rather than the operation that should have been ignored.

Workaround

Wait for STATUS == Ready before triggering a new task.

3.7 [16] VPR: Cleared EVENTS are triggered again on the next CNT0 event

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

TRIGGERED events are triggered again on the following CNT0 event.

Conditions

A TRIGGERED event is set with CSR and cleared.

Consequences

APB cannot be used to properly clear events if CNT0 is used.

Workaround

Set up a VPR TASK which triggers an interrupt in the VPR to clear the relevant VEVENTS bits through a CSR write. Do not use VPR EVENTS to trigger interrupts. Alternatively, use DPPI from the VPR to the EGU to interrupt the CPU.

This workaround is included in nRF Connect SDK version 3.0.0 and later.

3.8 [17] VPR: Bus faults during hibernate save sequence are ignored

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

Bus error responses during hibernate save are ignored.

Conditions

A bus fault occurs during the hibernate save sequence.

Consequences

The VPR CPU behavior is undefined.

Workaround

Before entering the lowest power state on VPR, ensure that the VPR context save memory register (VPRSAVEADDR) is set to memory address accessible to VPR.

3.9 [18] VPR: Subsequent VPR hibernate sleeps might fail to save ra

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

GPR x1 (ra) is not updated.

Conditions

VPR enters sleep in VPR hibernate mode multiple times without powering down.

Consequences

The VPR application might misbehave.

Workaround

After waking up from sleep in VPR hibernate mode, issue an APB write with VPR anywhere inside its own RTP or CPUCTRL peripheral address range.

3.10 [20] CLOCK: RADIO payload is not transmitted

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

Preamble is correctly transmitted but all following payloads are not transmitted.

Conditions

CPU and peripherals in MCU power domain are sleeping when RADIO starts transmitting.

Consequences

Transmission is corrupted.

Workaround

Before triggering RADIO.TASKS_RXEN or RADIO.TASKS_TXEN, trigger POWER.TASKS_CONSTLAT.

When disabling RADIO, trigger POWER.TASKS_LOWPWR.

The workaround might increase current consumption.

3.11 [21] SPIM: MOSI line toggles after transmission ends

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

One additional transition appears on the MOSI line at the end of the transmission when the clock has stopped toggling.

Conditions

SPIM00 or SPIM02 is being used with SPI mode 0. The first transmitted bit in the final byte is 1.

Consequences

None.

Workaround

None.

3.12 [22] TIMER: EVENTS_COMPARE[n] event is not generated

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

The generation of the EVENTS_COMPARE[n] does not occur.

Conditions

TIMER has been used in counter mode, counted to 1, and cleared. The power domain associated with TIMER is powered down. TIMER is then used in timer mode, and CC[n] is 1.

Consequences

The EVENT is missed.

Workaround

Apply one of the following workarounds:

- Do not use CC=1.
- Use SHORT register to stop and clear TIMER.
- Use PPI and START and CLEAR tasks at the same time.

3.13 [23] RRAMC: READY and READYNEXT events are generated when CPU wakes up

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

READY and READYNEXT are set when the CPU is waking up.

Conditions

The CPU wakes up.

Consequences

READY and READYNEXT cannot be relied on immediately after CPU wakes up.

Workaround

Do not use READY or READYNEXT in combination with CPU sleep.

3.14 [24] TIMER: EVENT_COMPARE is not generated when using PPI

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

The PPI enabled by PUBLISH_COMPARE[n] is not generated.

Conditions

DPPI channel is using TIMER EVENTS_COMPARE as an event with CC[n]=1.

Consequences

The PPI is missing.

Workaround

Apply one of the following workarounds:

- Do not use CC=1.
- Use SHORT register to stop and clear TIMER.
- Use PPI and START and CLEAR tasks at the same time.

3.15 [25] TIMER: EVENT_COMPARE[n] is not generated

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

EVENTS_COMPARE[n] is not generated for TIMER00 or TIMER10.

Conditions

The TIMER is configured with Timer mode and $CC[n]=x+1$, where x is the internal Counter value after TIMER is stopped. The TIMER is started once and then stopped with internal Counter at x. The power domain associated with the TIMER is powered down. The TIMER is started through PPI.

Consequences

The EVENT is missed.

Workaround

Apply one of the following workarounds for TIMER00 or TIMER10:

- Write $TIMER\ base + 0x010 = 1$ after STOP and before next START task.
- Trigger DPPI task at subscribe register $TIMER\ base + 0x090$ after stop and before next start.

This clears the counter.

Alternatively, the following workaround can be used for TIMER00:

Start timer with write to TASKS_START.

3.16 [26] DPPI: PPIB between domains uses wrong security attribution

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

PPIB channel security does not use the correct setting.

Conditions

DPPIC channels are configured with different S/NS security, and the PPIB channel subscribes to a DPPIC index different from the PPIB channel index.

Consequences

PPIB channels are blocked.

Workaround

Configure PPIB subscribe and publish to use the DPPIC channel of matching index, for example use DPPIC channel 0 for PPIB channel 0.

3.17 [30] CLOCK, GRTC: GRTC operates incorrectly at low temperature

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

GRTC drifts in frequency at low temperature.

Conditions

The system is using HFINT.

Consequences

Applications with real-time requirements can behave unexpectedly.

Workaround

A workaround is implemented in nRF Connect SDK 2.8.0 and later when MPSL is used.

A generic workaround is implemented in nRF Connect SDK 3.0.0 and later.

3.18 [37] POWER: Current consumption might increase after pin reset or power cycle

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

Current consumption is higher than expected in System OFF mode.

Conditions

The device is entering System OFF mode too soon after a pin reset or power cycle.

Consequences

Current consumption is higher than expected.

Workaround

Apply the following workaround:

1. Perform the following code: `*(volatile uint32_t *) 0x5005340C =1;`

2. Ensure that 40 CPU cycles are executed before entering System OFF mode.

3.19 [39] CLOCK: Device can behave erratically after XOSTART

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

The device behaves erratically or becomes unresponsive.

Conditions

The XOSTART task is triggered when the PLLSTART task is not triggered and CPU is sleeping.

Consequences

Peripherals outside the MCU domain can behave erratically.

Workaround

Trigger TASKS_PLLSTART before TASKS_XOSTART is triggered and trigger TASKS_PLLSTOP after TASKS_XOSTOP is triggered. Current consumption might increase depending on the use case.

3.20 [41] RADIO: Sensitivity is degraded on $n \times 32$ MHz channels

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

Sensitivity is reduced on the 2432 MHz and 2464 MHz channels.

Conditions

The radio is set to a receive channel which is an integer multiplier of the 32 MHz reference frequency.

Consequences

- Sensitivity is degraded on 1 Mbps BLE, 2 Mbps BLE, nRF1M, and nRF2M by approximately 2 dB.
- Sensitivity is degraded on nRF4M by approximately 2 dB on channels 2402, 2430, 2432, 2334, 2462, 2464, and 2466.
- Sensitivity is degraded on BLR125K by approximately 4 dB.
- Sensitivity is degraded on IEEE 802.15.4 by approximately 2 dB in the 2465 MHz RX frequency.

3.21 [44] EASYDMA: DMA.RX.AMOUNT is not updated with bytes count after match event

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

When a MATCH event is generated, DMA.RX.AMOUNT does not reflect the position of the character in memory.

Conditions

TWIM, SPI, or UART is used in RX mode with match candidates set up and match filter enabled.

Consequences

DMA.RX.AMOUNT cannot be used as a reliable offset with DMA.RX.PTR to find the matching character in memory.

3.22 [47] POWER: Device resets while debugging in System OFF mode

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

The device is not responsive to the debugger and resets during debug operations.

Conditions

The device is in emulated System OFF mode and at temperature above 98°C or has power-intensive operations in firmware immediately after the line putting it into System OFF mode.

Consequences

Debug access is unstable.

Workaround

After a failed attempt, retry debug operations.

3.23 [48] RESET: RESETREAS register might show an additional wakeup source after reset

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

RESE TREAS wakeup field has an incorrect value.

Conditions

A wakeup source is active during a reset.

Consequences

Reset reason might lead to an incorrect assumption on the application state.

Workaround

When RESE TREAS shows a pin reset (RESE TPIN), ignore other reset reason bits. Apply the following code after any reset:

```
if (NRF_RESET->RESE TREAS & RESET_RESE TREAS_RESE TPIN_Msk)
{
    NRF_RESET->RESE TREAS = ~RESET_RESE TREAS_RESE TPIN_Msk;
}
```

3.24 [49] RADIO: First bits of on-air packet are not correct

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

First bits of the PAYLOAD field in the RADIO on-air packet are not correct.

Conditions

PCNF0.S1LEN is 4 or less and PCNF0.S1INCL is 1. RADIO is transmitting using 1 Mbps, 2 Mbps, or 4 Mbps data rate. The MCU power domain is powered down.

Consequences

Transmission is incorrect.

Workaround

If S1LEN is not 0, apply the following workaround:

```
*(volatile uint32_t *) 0x5008C58C = 1;
```

Alternatively, use Constant Latency sub-power mode before starting the radio. If S1LEN is 0, apply the following workaround: Set S1 field equal to first byte in package.

3.25 [50] RRAMC: SPU00 incorrectly reports that RRAMC has DMA

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

SPU DMA capability register for RRAMC (SPU00->PERIPH[11].PERM, DMA field) reports NoSeparateAttribute, but RRAMC does not have DMA.

Conditions

SPU is used to find the DMA capabilities of the RRAM controller.

Consequences

The availability of a DMA is incorrectly detected.

Workaround

Do not use the DMA field of SPU00->PERIPH[11].PERM to find the capability of RRAMC.

3.26 [54] SPIS: SDO is configured as an input when SPIS is idle

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

Current consumption is higher than expected.

Conditions

The SPIS chip select (CSN) line is high, and the SDO line does not have pull-up or pull-down resistors.

Consequences

Current consumption increases by up to 0.5 mA when SPIS is used.

The additional current consumption is influenced by VDD and temperature.

Workaround

Ensure that the SDO has a known state and is not left floating.

For example, use an internal or external pull-down resistor on the SDO line.

3.27 [58] COMP: Static 300 μ A current after COMP has been enabled

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

Current consumption is increased when the device is in System ON all idle mode.

Conditions

The following conditions trigger failure mode:

- COMP state changes from ABOVE to BELOW to ABOVE to BELOW within 0.5 μ s.
- COMP state changes from BELOW to ABOVE to BELOW to ABOVE within 0.5 μ s.

It is recommended to enable hysteresis in COMP to reduce the risk of the described conditions occurring in noisy environments.

Consequences

Current consumption is approximately 300 μ A higher than specified in System ON all idle mode.

Workaround

Avoid the following conditions when COMP is enabled:

- COMP state changes from ABOVE to BELOW to ABOVE to BELOW within 0.5 μ s.
- COMP state changes from BELOW to ABOVE to BELOW to ABOVE within 0.5 μ s.

The anomaly is resolved by any COMP state changes from ABOVE to BELOW to ABOVE or from BELOW to ABOVE to BELOW where the time between the state changes is longer than 0.5 μ s.

3.28 [59] EGU: EGU uses incorrect security attribution

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

EGU interrupt uses the same security setting as DPPI channel 0 regardless of how it is configured.

Conditions

EGU is used, and the DPPI channel 0 security setting is different from the EGU peripheral's security setting.

Consequences

All interrupt-related registers use the same security setting as DPPI channel 0.

Workaround

Set the security attribution of the EGU peripheral to the desired level and apply the same security attribution to DPPI channel 0.

3.29 [63] RESET: Soft reset might not fully reset the device

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

The device does not behave as expected after a soft reset.

Conditions

A soft reset is triggered exactly at the same time as a power domain transitions from RUN state to IDLE state.

Consequences

Some parts of the device are not fully reset.

Workaround

Apply the appropriate workaround for the affected soft resets:

- Before AIRCR.SYSRESETREQ is used, apply constant latency submode.
- Prevent the processor from entering lockup state when code is executed in NMI or HardFault handlers.
- When CTRL-AP is used, avoid using soft reset and use pin reset instead.

3.30 [100] GPIO: Pins with analog functionality might be shorted together during power-on reset

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

During device startup, any of the following pins might be shorted together: P1.04 (AIN0), P1.05 (AIN1), P1.06 (AIN2), P1.07 (AIN3), P1.11 (AIN4), P1.12 (AIN5), P1.13 (AIN6), and P1.14 (AIN7).

During device startup, any of the following pins might be shorted to ground: P1.04 (AIN0), P1.06 (AIN2), P1.07 (AIN3), P1.11 (AIN4), P1.12 (AIN5), P1.09, and P1.10.

Pins shorting together and shorting to ground might occur simultaneously.

Conditions

The device is in power-on reset state. nRESET might prolong the time the pins are being shorted together if it is asserted during power-on reset.

Consequences

The following pins might be shorted together either for 10 μ s after power-on reset or while nRESET is asserted: P1.04 (AIN0), P1.05 (AIN1), P1.06 (AIN2), P1.07 (AIN3), P1.11 (AIN4), P1.12 (AIN5), P1.13 (AIN6), and P1.14 (AIN7).

The following pins might be shorted to ground for 1 μ s after power-on reset: P1.04 (AIN0), P1.06 (AIN2), P1.07 (AIN3), P1.11 (AIN4), P1.12 (AIN5), P1.09, and P1.10.

Workaround

Do not assert nRESET during power-on reset ($t_{POR,10\mu s}$), and ensure the application can tolerate shorted pins during power-on reset.

3.31 [102] CCM: Operation fails when CCM MAC length is set to 0

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

CCM might stop and not complete its operation when CONFIG.MACLEN=M0 is used.

Conditions

CCM is used for encryption without authentication and under some combinations of DMA address alignment.

Consequences

CCM stalls and does not generate the END or ERROR events.

Workaround

Set MODE.MACLEN=M4 and ignore the MAC written by CCM.

3.32 [103] GPIO: State is inconsistent during power-on reset

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

During device startup, the GPIO pins might not be in high-impedance state.

Conditions

Supply voltage drops below $V_{BOR,OFF}$ for a period of less than 50 ms.

Consequences

GPIO pins might be output, pull-down, or pull-up for a period of 10 μ s after power-on reset. If nRESET is asserted during power-on reset, GPIO pins might be output, pull-down, or pull-up until nRESET is deasserted.

Workaround

Ensure that supply voltage drops below $V_{BOR,OFF}$ are longer than 50 ms.

3.33 [105] TWIM: Peripheral becomes unresponsive when disabled during clock stretching

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

TWIM is unable to communicate with any peer device.

Conditions

TWIM is disabled during I2C transaction when clock stretching is ongoing.

Consequences

TWIM peripheral is unresponsive.

Workaround

Perform a soft reset or a hard reset of the device to recover the TWIM peripheral.

3.34 [111] SAADC: Output data is incorrect when noise shaping is used

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

SAADC output is incorrect, which can be observed as large intermittent errors or data inversion.

Conditions

SAADC is configured in single-ended mode, noise shaping is enabled, and input amplitude is over 50% of the ADC core's input range with the selected gain setting.

Consequences

SAADC output is incorrect.

Workaround

Apply one of the following workarounds:

- Use differential mode with negative input tied to GND.
- Use single-ended mode and guarantee that the input amplitude does not exceed 50% of the ADC core's input range with the selected gain setting.

3.35 [114] GPIO: Current consumption increases when Wake on pin is used

This anomaly applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous SoC revision [Revision 1](#).

Symptoms

Current consumption increases when the device is in System ON all idle mode.

Conditions

- GPIO.PIN_CFN.SENSE is enabled, and GPIO.PIN_CFN.INPUT is connected.
- DETECT signal changes from low to high to low to high within 0.5 μ s.
- SENSE condition remains active.

Consequences

Current consumption is approximately 300 μ A higher than specified in System ON all idle mode.

Workaround

Apply one of the following workarounds:

- Avoid the DETECT signal changing from low to high to low to high within 0.5 μ s.
- Enable latched mode by using GPIO:DETECTMODE = LDETECT.

4 Fixed anomalies

The anomalies listed in this table are no longer present in the current SoC version.

For a detailed description of the fixed anomalies, see the [Errata for Revision 1](#).

ID	Module	Description
31	POWER	Current consumption is higher than expected in sleep
33	RADIO	GPIOs can exhibit radiated emissions

Table 2: Fixed anomalies

5 New and inherited configurations

The following configurations apply to Revision 2 of the nRF54L15 SoC.

ID	Module	Description	Inherited from Revision 1
56	RADIO	Regulatory limits might be exceeded when using external FEM or PA	X

Table 3: Configurations

5.1 [56] RADIO: Regulatory limits might be exceeded when using external FEM or PA

This configuration applies to Revision 2, build codes QFAA-C00, CAAA-C00, QDAA-C00, QGAA-C00.

It was inherited from the previous IC revision [Revision 1](#).

Symptoms

When transmitting with an external RF front-end module (FEM) or power amplifier (PA), the power emitted in the 5-25 MHz frequency range might exceed regulatory limits.

Conditions

The device is operating with an external FEM or PA.

Consequences

The end product might fail regulatory conformance test.

Workaround

In Bluetooth and Nordic proprietary modes, use RADIO:TXPOWER lower than 4 dBm in the device.

In all modes, apply the following code before enabling the RADIO:

```
(volatile uint32_t *)(((uint32_t)NRF_RADIO) + 0x8C4) = 0x01280001ul;
```